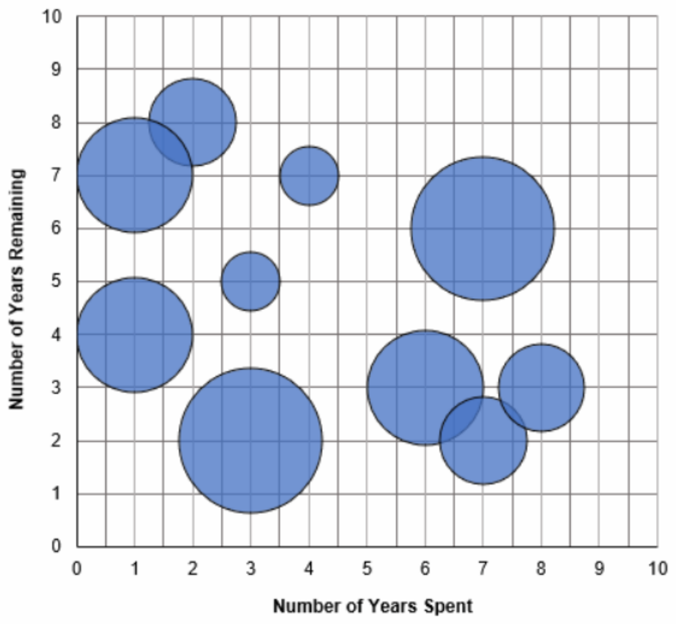
DIRECTIONS *for questions 1 to 5:* Answer the questions on the basis of the information given below.

At the beginning of 2022, Raju, a prison warden, was analysing the data on the prison sentences of the prisoners in his prison. For each prisoner, he collected information regarding the number of years that the prisoner had spent in the prison as part of his sentence and the number of years that he still has to spend in the prison to complete his sentence. He divided all the prisoners into groups, such that for all the prisoners in any group, the number of years spent in the prison was the same and the number of years left in the sentence was also the same. Based on this data, he made the chart given below.

In the chart, each bubble represents a group of prisoners, and the width of each bubble represents the number of prisoners in the group. The centre of each bubble represents the number of years spent in the prison (along the horizontal axis) and the number of years remaining in the sentence (along the vertical axis).

It is known that, for any prisoner, the number of years spent and the number of years remaining are both integers. Any prisoner commences his sentence at the beginning of a year and finishes his sentence at the end of a year.



Q1. DIRECTIONS *for question 1:* Select the correct alternative from the given choices.

Among the given prisoners, if exactly 1100 prisoners started their sentences after the end of 2016 and will finish their sentences before the beginning of 2027, what is the number of prisoners who will finish their sentences during the period starting from the beginning of 2023 to the end of 2027?

- a) 2300
- b) 2600
- c) 1800
- d) 2000

Q2. DIRECTIONS *for question 2:* Type in your answer in the input box provided below the question.

From 2014 to 2021, among the given prisoners, if, for exactly two consecutive years, the number of prisoners who started their sentences in the first year was 64 less than that of those who started their sentences in the second year, how many of the given prisoners will be in the prison at the beginning of 2026?

Q3. **DIRECTIONS** *for question 3:* Select the correct alternative from the given choices.

In 2023, exactly 100 prisoners were sent to Raju's prison and none of them are released in the same year. Further, from 2023 to 2029, at least 30 prisoners will be released at the end of each year and no other prisoner will be sentenced to prison from 2024 to 2030.

If, from 2023 to 2029, an equal number of prisoners will finish their sentences in each year and all the prisoners who were in the prison in 2022 or in 2023 will all be released by 2030, what is the number of prisoners who will finish their sentence at the end of 2030?

- a) 32
- b) 28
- c) 16
- d) Cannot be determined

Q4. DIRECTIONS *for question 4:* Type in your answer in the input box provided below the question.

In 2023, exactly 100 prisoners were sent to Raju's prison and none of them are released in the same year. Further, from 2023 to 2029, at least 30 prisoners will be released in each year and no other prisoner will be sentenced to prison from 2024 to 2030.

If, from 2023 to 2029, an equal number of prisoners will finish their sentences in each year and all the prisoners who were in the prison in 2022 or in 2023 will all be released by 2030, how many prisoners will be in the prison in June 2028?

Q5. DIRECTIONS *for question 5:* Select the correct alternative from the given choices.

Among the given prisoners, if the total number of prisoners whose duration of sentence is greater than 10 years is 40, how many of the given prisoners will finish their sentences in the year 2024?

- a) 28
- b) 42
- c) 63
- d) 35

DIRECTIONS *for questions 6 to 10:* Answer the questions on the basis of the information given below.

Ravi placed six cylindrical containers in a line, from left to right. The volume of each of the six containers is distinct among 100 ml, 250 ml, 350 ml, 600 ml, 900 ml and 1200 ml, not necessarily in any specific order. Exactly three of the six containers are empty, while the other three are filled to the brim with water. Ravi labelled these three filled containers as F1, F2 and F3, from left to right, in that order. Further, F1 is to the immediate right of an empty container and the last empty container from the right is to the immediate left of a filled container.

Ravi redistributed the water in the containers in the following manner:

- He took F1 and poured water from it into the first empty container from the right until F1 was half empty.
- He then took F2 and poured water from it into the second empty container from the right until F2 was half empty.
- He then took F3 and poured water from it into the third empty container from the right until F3 was half empty.

During this process, no water spilled out from any of the containers. Further, after Ravi finished redistributing the water between the six containers, the volumes of water in the first container from the left and the fourth container from the left are 300 ml and 450 ml, respectively. Also, the second container from the left has more quantity of water than the fifth container from the left.

Q6. DIRECTIONS *for questions 6 and 7:* Select the correct alternative from the given choices.
What is the total quantity of water (in ml) in all the containers after Ravi finished redistributing the water between the six containers?

a) 2700

b) 1750

c) 1850

d) 1600

DIRECTIONS *for questions 6 to 10:* Answer the questions on the basis of the information given below.

Ravi placed six cylindrical containers in a line, from left to right. The volume of each of the six containers is distinct among 100 ml, 250 ml, 350 ml, 600 ml, 900 ml and 1200 ml, not necessarily in any specific order. Exactly three of the six containers are empty, while the other three are filled to the brim with water. Ravi labelled these three filled containers as F1, F2 and F3, from left to right, in that order. Further, F1 is to the immediate right of an empty container and the last empty container from the right is to the immediate left of a filled container.

Ravi redistributed the water in the containers in the following manner:

- He took F1 and poured water from it into the first empty container from the right until F1 was half empty.
- He then took F2 and poured water from it into the second empty container from the right until F2 was half empty.
- He then took F3 and poured water from it into the third empty container from the right until F3 was half empty.

During this process, no water spilled out from any of the containers. Further, after Ravi finished redistributing the water between the six containers, the volumes of water in the first container from the left and the fourth container from the left are 300 ml and 450 ml, respectively. Also, the second container from the left has more quantity of water than the fifth container from the left.

Q7. DIRECTIONS *for questions 6 and 7:* Select the correct alternative from the given choices.
What is the capacity (in ml) of the third container from the left?

a) 100

b) 250

c) 350

d) Cannot be determined

DIRECTIONS *for questions 6 to 10:* Answer the questions on the basis of the information given below.

Ravi placed six cylindrical containers in a line, from left to right. The volume of each of the six containers is distinct among 100 ml, 250 ml, 350 ml, 600 ml, 900 ml and 1200 ml, not necessarily in any specific order. Exactly three of the six containers are empty, while the other three are filled to the brim with water. Ravi labelled these three filled containers as F1, F2 and F3, from left to right, in that order. Further, F1 is to the immediate right of an empty container and the last empty container from the right is to the immediate left of a filled container.

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- He then took F3 and poured water from it into the third empty container from the right until F3 was half empty.

During this process, no water spilled out from any of the containers. Further, after Ravi finished redistributing the water between the six containers, the volumes of water in the first container from the left and the fourth container from the left are 300 ml and 450 ml, respectively. Also, the second container from the left has more quantity of water than the fifth container from the left.

Q8. **DIRECTIONS** *for questions 8 and 9:* Type in your answer in the input box provided below the question.

What is the total quantity (in ml) of water that is present in the second, fourth and fifth containers from the left, after Ravi finished redistributing the water between the six containers?

DIRECTIONS *for questions 6 to 10:* Answer the questions on the basis of the information given below.

Ravi placed six cylindrical containers in a line, from left to right. The volume of each of the six containers is distinct among 100 ml, 250 ml, 350 ml, 600 ml, 900 ml and 1200 ml, not necessarily in any specific order. Exactly three of the six containers are empty, while the other three are filled to the brim with water. Ravi labelled these three filled containers as F1, F2 and F3, from left to right, in that order. Further, F1 is to the immediate right of an empty container and the last empty container from the right is to the immediate left of a filled container.

Ravi redistributed the water in the containers in the following manner:

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- He then took F3 and poured water from it into the third empty container from the right until F3 was half empty.

During this process, no water spilled out from any of the containers. Further, after Ravi finished redistributing the water between the six containers, the volumes of water in the first container from the left and the fourth container from the left are 300 ml and 450 ml, respectively. Also, the second container from the left has more quantity of water than the fifth container from the left.

Q9. **DIRECTIONS** *for questions 8 and 9:* Type in your answer in the input box provided below the question.
What is the maximum quantity (in ml) of water that can be further poured into any of the six containers, after Ravi finished redistributing the water between the six containers?

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DIRECTIONS *for questions 6 to 10:* Answer the questions on the basis of the information given below.
Ravi placed six cylindrical containers in a line, from left to right. The volume of each of the six containers is distinct among 100 ml, 250 ml, 350 ml, 600 ml, 900 ml and 1200 ml, not necessarily in any specific order. Exactly three of the six containers are empty, while the other three are filled to the brim with water. Ravi labelled these three filled containers as F1, F2 and F3, from left to right, in that order. Further, F1 is to the immediate right of an empty container and the last empty container from the right is to the immediate left of a filled container.

Ravi redistributed the water in the containers in the following manner:

- He took F1 and poured water from it into the first empty container from the right until F1 was half empty.
- He then took F2 and poured water from it into the second empty container from the right until F2 was half empty.
- He then took F3 and poured water from it into the third empty container from the right until F3 was half empty.

During this process, no water spilled out from any of the containers. Further, after Ravi finished redistributing the water between the six containers, the volumes of water in the first container from the left and the fourth container from the left are 300 ml and 450 ml, respectively. Also, the second container from the left has more quantity of water than the fifth container from the left.

Q10. **DIRECTIONS** *for question 10:* Select the correct alternative from the given choices.
Ravi numbered the six containers from 1 to 6, starting from left to right, in that order. If Ravi emptied all the water from container 1 into container 2, and then from container 3 into container 4 and, finally, from container 5 into container 6, from how many containers would the water have spilled over?

a) 0

b) 1

c) 2

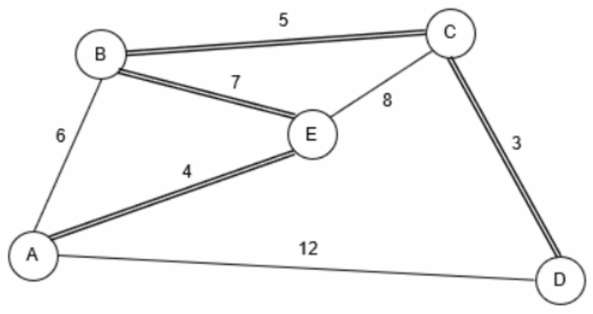
d) 3

DIRECTIONS *for questions 11 to 15:* Answer the questions on the basis of the information given below.

Five islands – A, B, C, D and E – are connected by a certain number of bridges, as shown in the figure below. Any person who wants to travel from one island to another has to use these bridges.

However, all the bridges are rickety and will collapse after a certain number of persons walk on them (after a bridge collapses, no one else can walk on that bridge). In the figure given below, the bridges represented by a single line will collapse after exactly 25 persons walk on the bridge, while the bridges represented by a double line will collapse after exactly 50 persons walk on the bridge. Also, the numbers given along each bridge represents the time taken (in minutes) by any person to cross the bridge. Further, on any bridge, assume that any number of persons can walk on the bridge starting at the same time.

Any group of persons who want to travel from an origin island to a destination island know the nature of each bridge (i.e., that the bridge will collapse after a certain number of persons walk on the bridge) and will plan to minimize the total time taken for the group to travel from one island to another, based on the size of the group and the nature of the bridges. The time taken for a group to travel from an origin island to a destination island is the difference between the time at which the first person starts at the origin and the time at which the last person reaches the destination.



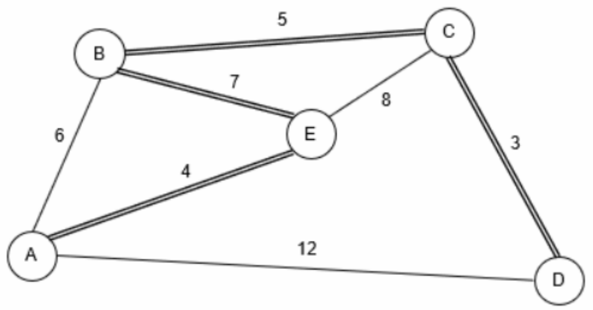
Q11. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.
What is the maximum number of persons who can travel from A to E?

- a) 50
- b) 75
- c) 100
- d) 125

DIRECTIONS *for questions 11 to 15:* Answer the questions on the basis of the information given below.
Five islands – A, B, C, D and E – are connected by a certain number of bridges, as shown in the figure below. Any person who wants to travel from one island to another has to use these bridges.

However, all the bridges are rickety and will collapse after a certain number of persons walk on them (after a bridge collapses, no one else can walk on that bridge). In the figure given below, the bridges represented by a single line will collapse after exactly 25 persons walk on the bridge, while the bridges represented by a double line will collapse after exactly 50 persons walk on the bridge. Also, the numbers given along each bridge represents the time taken (in minutes) by any person to cross the bridge. Further, on any bridge, assume that any number of persons can walk on the bridge starting at the same time.

Any group of persons who want to travel from an origin island to a destination island know the nature of each bridge (i.e., that the bridge will collapse after a certain number of persons walk on the bridge) and will plan to minimize the total time taken for the group to travel from one island to another, based on the size of the group and the nature of the bridges. The time taken for a group to travel from an origin island to a destination island is the difference between the time at which the first person starts at the origin and the time at which the last person reaches the destination.



Q12. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.
A group of 50 persons are planning to travel from D to E. What is the minimum time (in minutes) taken by the group to travel from D to E?

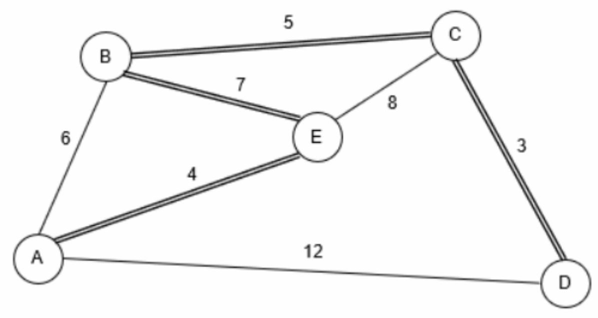
- a) 11
- b) 12
- c) 15
- d) 16

DIRECTIONS *for questions 11 to 15:* Answer the questions on the basis of the information given below.

Five islands – A, B, C, D and E – are connected by a certain number of bridges, as shown in the figure below. Any person who wants to travel from one island to another has to use these bridges.

However, all the bridges are rickety and will collapse after a certain number of persons walk on them (after a bridge collapses, no one else can walk on that bridge). In the figure given below, the bridges represented by a single line will collapse after exactly 25 persons walk on the bridge, while the bridges represented by a double line will collapse after exactly 50 persons walk on the bridge. Also, the numbers given along each bridge represents the time taken (in minutes) by any person to cross the bridge. Further, on any bridge, assume that any number of persons can walk on the bridge starting at the same time.

Any group of persons who want to travel from an origin island to a destination island know the nature of each bridge (i.e., that the bridge will collapse after a certain number of persons walk on the bridge) and will plan to minimize the total time taken for the group to travel from one island to another, based on the size of the group and the nature of the bridges. The time taken for a group to travel from an origin island to a destination island is the difference between the time at which the first person starts at the origin and the time at which the last person reaches the destination.



Q13. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.

A group of persons want to travel from B to C and the minimum time taken (in minutes) by the group to reach their destination is m. What is the maximum possible value of m?

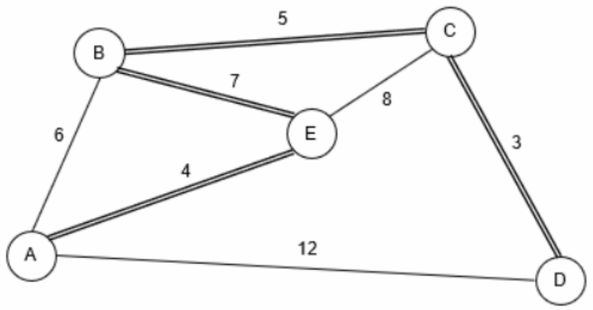
- a) 18
- b) 21
- c) 23
- d) 26

DIRECTIONS *for questions 11 to 15:* Answer the questions on the basis of the information given below.

Five islands – A, B, C, D and E – are connected by a certain number of bridges, as shown in the figure below. Any person who wants to travel from one island to another has to use these bridges.

However, all the bridges are rickety and will collapse after a certain number of persons walk on them (after a bridge collapses, no one else can walk on that bridge). In the figure given below, the bridges represented by a single line will collapse after exactly 25 persons walk on the bridge, while the bridges represented by a double line will collapse after exactly 50 persons walk on the bridge. Also, the numbers given along each bridge represents the time taken (in minutes) by any person to cross the bridge. Further, on any bridge, assume that any number of persons can walk on the bridge starting at the same time.

Any group of persons who want to travel from an origin island to a destination island know the nature of each bridge (i.e., that the bridge will collapse after a certain number of persons walk on the bridge) and will plan to minimize the total time taken for the group to travel from one island to another, based on the size of the group and the nature of the bridges. The time taken for a group to travel from an origin island to a destination island is the difference between the time at which the first person starts at the origin and the time at which the last person reaches the destination.



Q14. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.

If a group of 75 persons took more than 15 minutes to travel from D to another island, say X, how many possibilities exist for X?

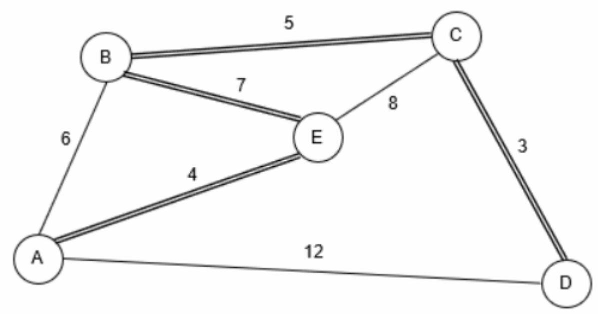
- a) 1
- b) 4
- c) 2
- d) 3

DIRECTIONS *for questions 11 to 15:* Answer the questions on the basis of the information given below.

Five islands – A, B, C, D and E – are connected by a certain number of bridges, as shown in the figure below. Any person who wants to travel from one island to another has to use these bridges.

However, all the bridges are rickety and will collapse after a certain number of persons walk on them (after a bridge collapses, no one else can walk on that bridge). In the figure given below, the bridges represented by a single line will collapse after exactly 25 persons walk on the bridge, while the bridges represented by a double line will collapse after exactly 50 persons walk on the bridge. Also, the numbers given along each bridge represents the time taken (in minutes) by any person to cross the bridge. Further, on any bridge, assume that any number of persons can walk on the bridge starting at the same time.

Any group of persons who want to travel from an origin island to a destination island know the nature of each bridge (i.e., that the bridge will collapse after a certain number of persons walk on the bridge) and will plan to minimize the total time taken for the group to travel from one island to another, based on the size of the group and the nature of the bridges. The time taken for a group to travel from an origin island to a destination island is the difference between the time at which the first person starts at the origin and the time at which the last person reaches the destination.



Q15. DIRECTIONS *for questions 11 to 15:* Select the correct alternative from the given choices.
If a group with the maximum possible number of persons travelled from B to C, what is the minimum time (in minutes) taken by the group to travel from B to C?

- a) 15
- b) 28
- c) 21
- d) 26

DIRECTIONS *for questions 16 to 20:* Answer the questions on the basis of the information given below.

Five friends, A through E, weighed themselves at the beginning of January, and found their weights to be distinct among 120 kg, 130 kg, 150 kg, 180 kg and 200 kg, not necessarily in the same order. However, starting from January till April, each of the five friends followed a strict regimen of diet and exercise, because of which each friend lost a fixed amount of weight every month. The weight (in kg) that each friend lost in each month was a distinct integer among 3, 5, 7, 10 and 14, not necessarily in any order.

At the end of each month, from January to April, the five friends ranked themselves, from 1 to 5, in the descending order of their weights (i.e., the heaviest was ranked 1, while the lightest was ranked 5).

It is also known that

i. the weight of A at the end of February was a multiple of 8, but he was not the person who lost 5 kg every month.

ii. the weight of C was greater than that of D in both March and April but not in February.

iii. the person who was ranked first at the end of January weighed 67 kg more than the person who was ranked fourth at the end of January.

iv. the person who weighed 150 kg at the beginning of January did not lose 3 kg every month.

v. E was not ranked second at the end of April.

Q16. **DIRECTIONS** *for question 16:* Select the correct alternative from the given choices.

How many persons weighed at least 110 kg at the end of April?

- a) 2
- b) **3** ✓ **Your answer is correct**
- c) 4
- d) 5

Q17. DIRECTIONS *for question 17:* Type in your answer in the input box provided below the question.

At the end of April, what was the difference in weight (in kg) between the person who was ranked second and D?

Q18. DIRECTIONS *for question 18:* Select the correct alternative from the given choices.

For how many persons was their rank at the end of January the same as that at the end of April?

- a) 0
- b) 1 ✔ **Your answer is correct**
- c) 2
- d) 3

Q19. DIRECTIONS *for question 19:* Type in your answer in the input box provided below the question.

What was the weight (in kg) of C at the end of April?

Q20. DIRECTIONS *for question 20:* Select the correct alternative from the given choices.
The weight of which of the following persons was exactly 150 kg at the beginning of January?

a) E

b) **D**

c) A

d) **B** ✓Your answer is correct